

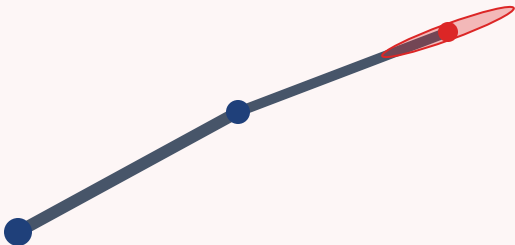
KINEMATIC REACHABILITY & MANIPULABILITY FILTERING

Filtering VLM Semantic Goals Against Yoshikawa Manipulability Index $w = \sqrt{(\det(J(q) J(q)^T))}$

× REJECTED: SINGULAR / OUT-OF-REACH

Manipulability Index $w < 0.05$ (Near Gimbal Lock)

Flattened Velocity Ellipsoid



- Arm fully outstretched at joint limit
- Inverse Kinematics Jacobian $\det(J) \rightarrow 0$
- Joint velocities explode $\dot{q} = J^{-1} v \rightarrow \infty$
- **Immediate Intent Rejection & Back-Off**

✓ ACCEPTED: HIGH ISOTROPY GOAL

Manipulability Index $w > 0.45$ (Condition Number $\kappa \approx 1.2$)

Isotropic Velocity Ellipsoid



- Well within reachable dextrous workspace
- Uniform control authority in all Cartesian directions
- Bounded motor torques $\tau = J^T F_{ext}$
- **Intent Certified & Dispatched to Planner**